

NATIONAL INSTITUTE OF TECHNOLOGY DURGAPUR

Odd Semester Mid-Term Examination, 2024-'25

Course Code: CSC 301
Course Name: Discrete Mathematics

Full Marks: 25
Time: 1.5 Hours

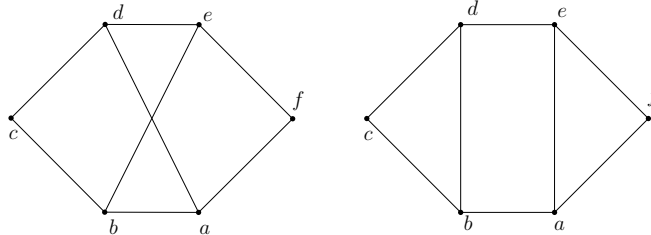
- Instructions:**
- i. Answer **five** questions taking **at least 2(two)** questions from each **Section**.
 - ii. Any missing value can be assumed suitably.

Section A

1. Show that in a group of 10 people (where any two people are either friends or enemies), there are either three mutual friends or four mutual enemies, and there are either three mutual enemies or four mutual friends. Use this to show that among any group of 20 people (where any two people are either friends or enemies), there are either four mutual friends or four mutual enemies. 5 CO3
2. [a] Let R be an equivalence relation defined on a set A . Show that every partition of A gives rise to a unique equivalence relation on A and every equivalence relation R defined on A results in a unique partition of A . 3 + 2 CO3, CO4
2. [b] Count the maximum possible number of reflexive relations and symmetric relations that can be defined on a set of n elements.
3. Let A be a set, and f, g, h three functions $A \rightarrow A$ such that $h \circ g \circ f$ is the identity function 1_A of A . Assume that neither of f, g, h is the identity function of A . Prove that f must be injective (one-one), and h must be surjective (onto). 5 CO2, CO3, CO4
[An identity function is a function where each element in a set A gives the image of itself as the same element i.e., $f(a) = a, \forall a \in A$.]
4. [a] State and prove the De Morgan's law of set theory. 3 + 2 CO1, CO2
4. [b] If $|A| = n$, then what is the value of $|Power_Set(A)|$? Prove it.

Section B

5. [a] Prove that in a graph $G = (V, E)$ the number of odd vertices is even. 2 + 3 CO2, CO5
5. [b] The complement of a graph $G = (V, E)$, denoted as $\overline{G}$, is defined such that it shares the same set of vertices as G , but its edges connect vertices that are not directly connected in G . Show that for any graph with $|V| = 6$, G or $\overline{G}$ contains a triangle.
6. [a] Check whether the following pair of graphs are isomorphic or not. Explain. 2 CO1, CO5



6. [b] Define tree. Derive the number of pendent vertices of a binary tree with n vertices. 1 + 2
7. [a] Apply (i) Euler formula or related corollary and (ii) Kuratowski Theorem - to show that K_6 is non-planar. 4 + 1 CO2, CO3
7. [b] Find a chromatic partition of the following graph.

